

ABSTRACT

DEXTROUS represents the pinnacle of collaborative innovation in robotics, driven by a shared passion for technological advancement. Our project centers around the development of a sophisticated 6-degree-of-freedom (6-DOF) robotic arm, poised to redefine automation technology. The project team, comprising diverse experts in robotics and engineering, strategically selected the ESP32 and Raspberry Pi 3 platforms for their unparalleled versatility and adaptability to robotics applications.

At its core, DEXTROUS embodies a dual commitment to innovation and skill enhancement. Drawing upon our collective expertise in mechanical design, electronics, and programming, we navigate each challenge with unwavering determination and a spirit of resilience. Each obstacle encountered serves not as a roadblock but as a catalyst for personal and collective growth, pushing the boundaries of our technical prowess.

DEXTROUS symbolizes the fusion of passion, expertise, and ambition, creating a collaborative environment where exploration and discovery flourish. Through our combined efforts, we aim to transcend existing limitations and create a groundbreaking 6-DOF robotic arm that epitomizes the intersection of challenge and innovation.

In summary, DEXTROUS stands as a testament to our relentless pursuit of excellence in robotics. It is more than a project; it is a manifestation of our collective vision to revolutionize automation technology and shape the future of robotics through ingenuity, collaboration, and unwavering determination.

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List of figures:

Sr. No	Figure number and Figure caption	Page no.
1	Fig. 2.1. Block schematic of system	9
2	Fig.2.2. Detailed block schematic of the system	10
3	Fig.4.1. CAD Diagram	20
4	Fig.4.2. Schematics of PCB	21
5	Fig.4.3. PCB Layout	21
6	Fig.4.4. Top View of PCB	21
7	Fig.4.5. Side View of PCB	21
8	Fig.4.6. Pairing of PS4	22
9	Fig.4.7. Entire Robot in Box for Size Mapping	22
10	Fig.4.8. Locomotion Testing	22
11	Fig.4.9. Bot's Half Arm Testing	22
12	Fig.4.10. Completed Robot with Arm	22
13	Fig.4.11. Complete Bot in Working State	22
14	Fig.5.1. BER Performance of Basic Linear Array	23

List of tables:

Sr. No	Table number and Table caption	Page no.
1	Table 2.1. : Transistor selection table	12
2	Table 3.1. : List of components required in the project	16
3	Table 4.1. : Tests and Consecutive Results Table	18
4	Table 5.1. : Bill of Materials Table	24

CONTENTS

	Abstract	i
	List of Figures	ii
	List of Tables	iii
	Abbreviations	iv
1	Introduction	2-7
	1.1 Background and Context	2
	1.2 Relevance	2
	1.3 Literature Survey	3
	1.4 Motivation	4
	1.5 Aim and Objectives	5
	1.6 Technical Approach	6
2	Block Diagram	7-11
	2.1 Theoretical background	7
	2.2 Block diagram	7
	2.3 Requirements	9
	2.4 Selection of Sensors and Major Components	10
3	System Design	11-16
4	Implementation, Testing and Debugging	16-21
5	Results and discussion	21
	Conclusions and Future Scope	22-23
	Bill of Material	24
	References	25
	Datasheets	26-44

Feasibility report

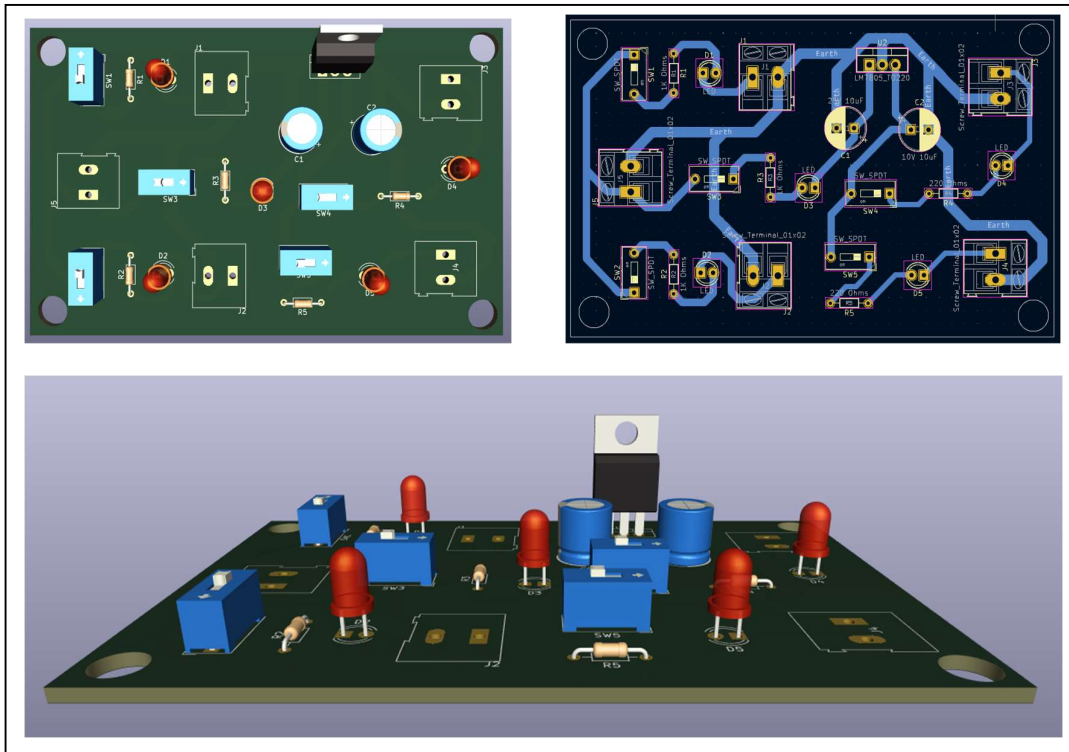
Title : DEXTROUS - 6 DOF Robotic Arm

Group members:

1) Vishal Sanjay Bhokare 2) Vedant Dinesh Butala 3) Sumedh Ravindra Kasat

Tools required	Testing possibility	Controller	Cost
Hardware/components	Hardware Yes	ESP32 PCA9685 MG995/996 LM2596S DC-DC Buck	Rs. 1648 /-
Software	Software Yes	1) Fusion 360 2) KiCAD 3) Arduino IDE	Free with Student License
Tools available within campus or outside	Sensors required	Signal conditioning if any	-
1) Blue Wire Cutter 2) Electric Soldering Tool 3) Breadboard 4) Multimeter 5) Wires and Jumper Cables 6) Power Tools in Workshop 7) CAD Lab for 3D Printing	The robotic arm employs key sensors for accurate operation: position encoders, proximity sensors, IMUs, current sensors, temperature sensors, and optional force sensors. Additionally, voltage/current and communication interface sensors enhance power management and data exchange efficiency.	Signal conditioning components enhance sensor performance by amplifying, filtering, converting, linearizing, calibrating, shifting voltage levels, isolating, and integrating signals for improved accuracy.	-
Applications if any	PCB design and fabrication	Datasheets/ application notes available	-
The robotic arm finds applications in various fields such as manufacturing, automation, logistics, healthcare, and research. Its versatility allows it to perform tasks like assembly, pick-and-place operations, material handling, surgical assistance, and experimentation in controlled environments.	- KiCad EDA 8.0.1 - Flux.ai	Datasheets and application notes are essential references providing detailed specifications, characteristics, and usage guidelines for project components. - MG995/996 - ESP32 - PS4 Controller - PCA9685 Module	-
Mechanical design	Enclosure design	Demonstration	-
The mechanical design encompasses the physical structure and components of the robotic arm, ensuring robustness, precision, and functionality.	The enclosure design involves the creation of a protective casing or housing for the robotic arm components, safeguarding them from environmental factors and ensuring safe operation.	The demonstration involves highlighting the robotic arm's functionality, including its motion capabilities, interaction with external devices, and execution of predefined tasks.	-

Title of project : DEXTROUS - 6 DOF Robotic Arm



Electric specification

Input Specifications:

- Voltage Range: 12V DC
- Current Rating: 2A

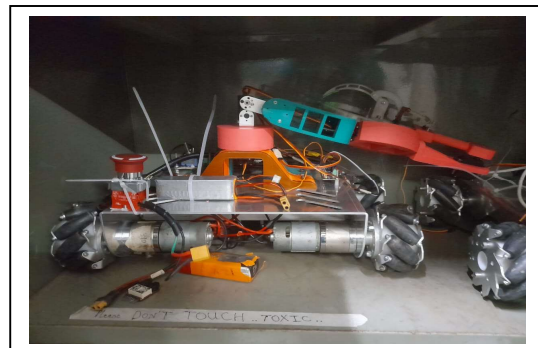
Output Specifications:

- Voltage: 5V DC
- Current: Up to 1A

Features:

- Overcurrent & Short Circuit Protection
- Efficiency: >60%

Mechanical specification



High Efficiency, Compact Design, and Robust Protection Features.
Cost of System : Rs. 27,035

The cost of the system is competitively priced, offering exceptional value for its efficiency, reliability, and features.

Group members:

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